

Decision Tree Model for Predicting Titanic Survival

Darren Easler

For 891 passengers of the Titanic, their information and survival data trained a Decision Tree machine learning model to predict survivors based on some of their biographic features. The features used for the model training are the following.

```
['Pclass', 'Sex', 'Age', 'SibSp', 'Parch', 'Fare', 'Embarked']
```

Four features were removed from this analysis, PassengerId (no contribution to survival, just identifier), Name (no contribution to survival, just identifier), Cabin (tons of missing data, does not predict survival well), and Ticket (no contribution to survival, just identifier). The seven features listed above are the most appropriate for training the model.

After removing unnecessary features, there were still two columns with missing data, Age and Embarked. The Age column was filled with the median age from the data. Only two data rows were missing Embarked values, so these were filled with the mode value. After these changes, there were no missing values in the cleaned dataset.

Of the remaining features, two were still categorical and were changed to numeric or boolean via `pd.get_dummies()`. These features were Sex and Embarked, which were changed to booleans, making new feature categories Sex_Male (0/1) and booleans Embraked_Q and Embarked_S.

The dataset was split using the typical 80/20 train/test split. The Decision Tree model was fit with the training data, and then predictions were made with the model and the testing feature data. The model results produced the following confusion matrix and evaluation metrics:

SURVIVAL CONFUSION MATRIX	Predicted 0	Predicted 1
Actual 0	84	21
Actual 1	18	56

Accuracy: 0.782
Precision: 0.727
Recall: 0.757

The model is accurate in classification 78.2% of the time. The precision of 72.7% tells that the positive predictions are correct, or that 27% of positive indications will be false. Lastly, the recall of 75.7% says that 24% of unflagged data was missed by the model.

In terms of the important features in the model, the most significant determining features of survival or not are overwhelmingly Sex_Male, Fare, and Age. Gender, Class, and Age are common denominators when it comes to who to save on the Titanic. Each feature is show with its relative importance below.

